

# Puzzle sheet

## Trigonometry equations

|      |      |
|------|------|
|      |      |
| 15.6 | 15.7 |

|      |      |      |      |      |      |      |      |  |  |
|------|------|------|------|------|------|------|------|--|--|
|      |      |      |      |      |      |      |      |  |  |
| 14.2 | 16.4 | 48.2 | 43.1 | 37.7 | 15.6 | 16.4 | 16.1 |  |  |

|      |      |      |
|------|------|------|
|      |      |      |
| 22.4 | 14.2 | 16.4 |

|      |      |
|------|------|
|      |      |
| 16.1 | 57.8 |

|      |      |      |      |      |  |
|------|------|------|------|------|--|
|      |      |      |      |      |  |
| 12.7 | 26.4 | 57.8 | 16.4 | 16.1 |  |

|      |      |
|------|------|
|      |      |
| 15.6 | 43.1 |

|      |      |      |      |
|------|------|------|------|
|      |      |      |      |
| 12.7 | 15.6 | 35.6 | 35.6 |

The answer to each question below has a corresponding letter. The numbers in the grid above are the answers to the questions. Determine the values of the letters in the questions, to one decimal place. Then fill in the boxes above with the letters that match the answers, to discover a well-known saying (often referred to as Murphy's Law).

### Questions

1  $\sin(52^\circ) = \frac{A}{18}$     **A** = \_\_\_\_\_

2  $\cos(47^\circ) = \frac{N}{24}$     **N** = \_\_\_\_\_

3  $\tan(18^\circ) = \frac{W}{39}$     **W** = \_\_\_\_\_

4  $\cos(63^\circ) = \frac{12}{R}$     **R** = \_\_\_\_\_

5  $\sin(19.4^\circ) = \frac{16}{Y}$     **Y** = \_\_\_\_\_

6  $\tan(26.5^\circ) = \frac{C}{45}$     **C** = \_\_\_\_\_

7  $\tan(39^\circ) = \frac{13}{G}$     **G** = \_\_\_\_\_

8  $\cos(54^\circ) = \frac{34}{O}$     **O** = \_\_\_\_\_

9  $\sin(11^\circ) = \frac{6.8}{L}$     **L** = \_\_\_\_\_

10  $\cos(65^\circ) = \frac{I}{37}$     **I** = \_\_\_\_\_

11  $\tan(78^\circ) = \frac{74}{F}$     **F** = \_\_\_\_\_

12  $\cos(61.5^\circ) = \frac{18}{H}$     **H** = \_\_\_\_\_

13  $\sin(26.4^\circ) = \frac{T}{97}$     **T** = \_\_\_\_\_